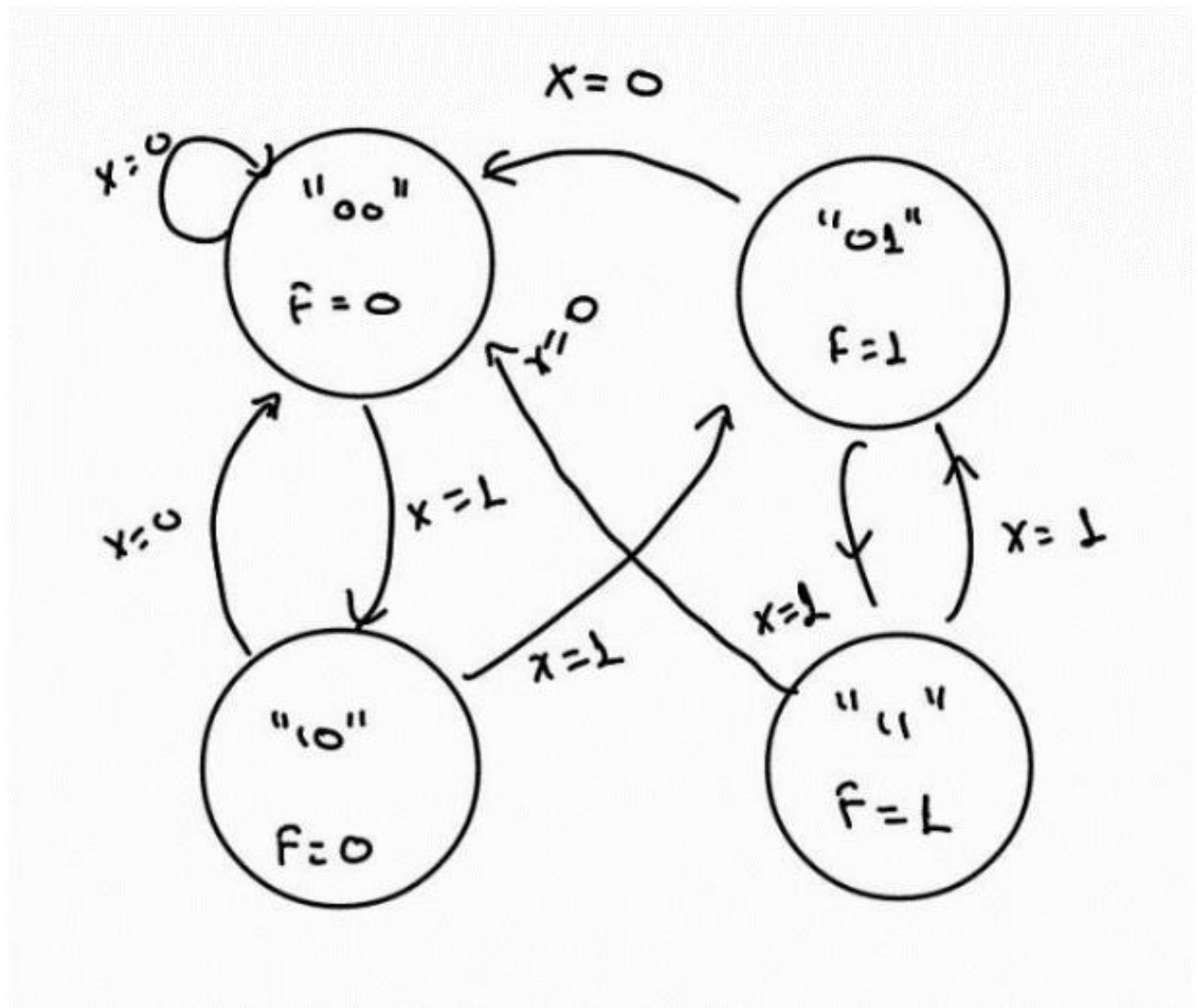


Lab 7: Finite State Machine

In this lab, we were expected to design FSM. By using one 2-input AND gate(74HC08), one 2-input OR gate(74HC32) and a D flip-flop(74HC74) the design is created. The D flip-flops outputs are connected via using the AND gate so that the output F is obtained.

The state diagram shown below:



(Figure 1: State diagram of the design)

State table of this design is shown below:

Current State " Q_0Q_1 "	Next State		Output
	$X = 0$	$X = 1$	
00	00	10	0
01	00	11	1
10	00	01	0
11	00	01	0

(Figure2: State table of design)

Karnaugh map for the output F is shown below:

		Q_0	
		0	1
Q_1	0	0	1
	1	0	0

(Figure3: Karnaugh map for output F)

Karnaugh map for flip-flop D_0 is shown below:

X	Q_0Q_1			
	00	01	11	10
0	0	0	0	0
1	0	1	1	1

(Figure4: Karnaugh map for D_0)

Karnaugh map for flip-flop D_1 is shown below:

$x \backslash Q_0Q_1$	00	01	11	10
0	0	0	0	0
1	1	1	0	0

(Figurer5: Karnaugh map for D_1)

By inspecting the karnaugh maps, we can arrive to the equations:

$$F = Q_1' * Q_0$$

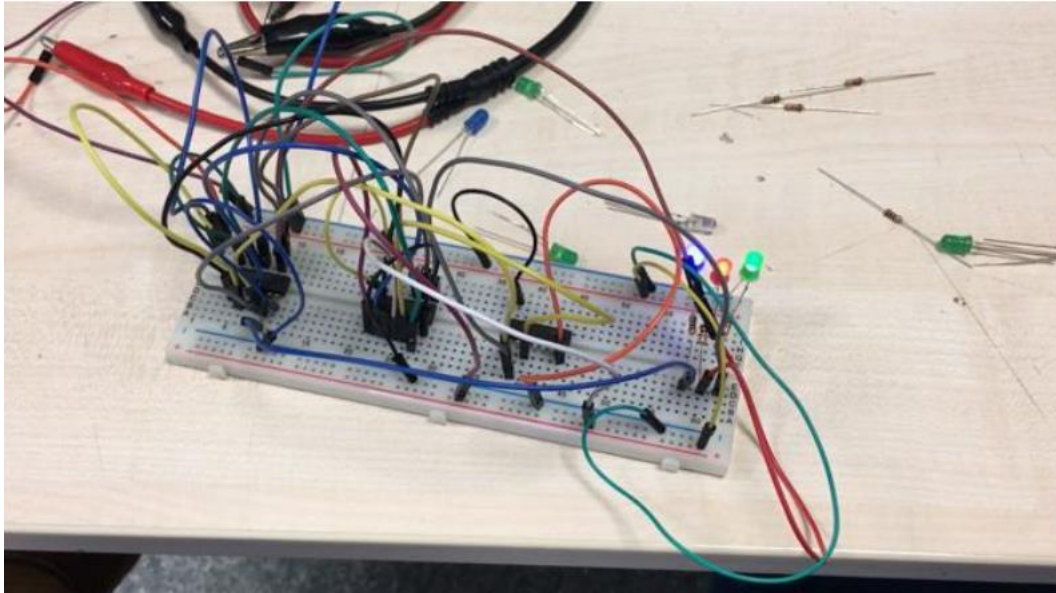
$$D_0 = (Q_0 + Q_1) * X$$

$$D_1 = Q_1' * X$$

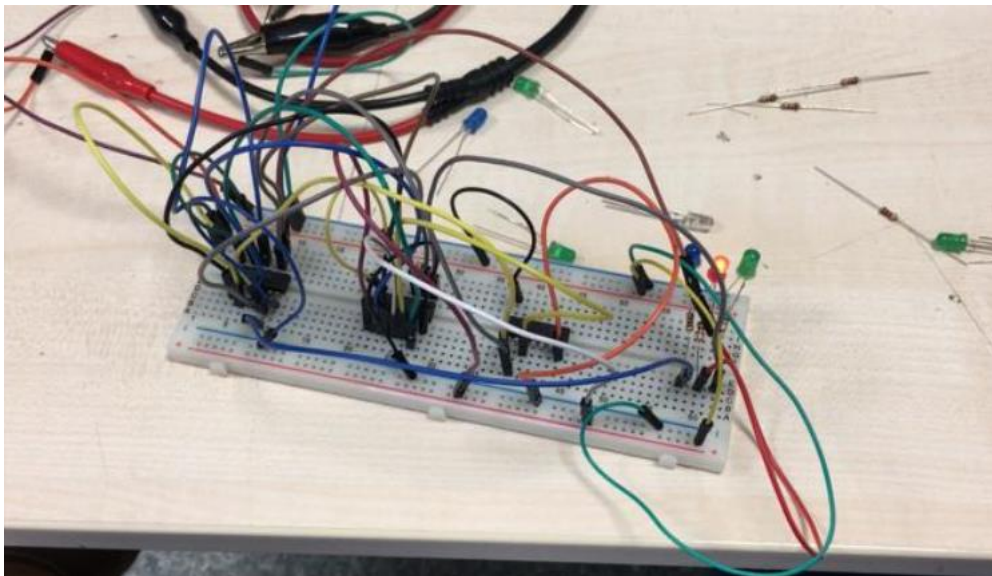
Process

I designed my finite state machine and drew the state diagram, state table, and karnaugh maps in the first portion of the experiment. The circuit equations were then derived using the karnaugh maps. I drew the circuit design schematic using the formulae. According to my design schematic, I created my circuit using a 2-input AND gate, a 2-input OR gate, and a dual D flip-flop in the second half of the lab. I then inspected the outputs using the signal generator as a clock and the power supply.

Results and Observations



(Figure 6: $Q = 11$ and $F = 1$)



(Figure 7: $Q = 01$ and $F = 0$)